

MARITIME BORDER ALERT SYSTEM

A PROJECT REPORT

submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled “**MARITIME BORDER ALERT SYSTEM**” is the bonafide work of “**LOKESH SINGH N(230701166),KIRUTHIKA KM(230701153),MONISH G(230701196)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The increasing instances of Rameshwaram fishermen unintentionally crossing the International Marine Boundary Line (IMBL) into Sri Lankan waters have resulted in serious consequences, including arrests and loss of livelihood. This project proposes a LoRa-based mobile application designed to address this critical issue by providing real-time alerts and accurate boundary visualization. Leveraging GPS technology and long-range, low-power LoRa communication, the app detects a user's proximity to the IMBL and issues timely visual and audio warnings, even in low-connectivity maritime zones. Built with empathy towards the daily challenges and technological limitations of local fishermen, the app features a user-friendly interface and offline map support. This solution aims to empower fishermen with the tools necessary to ensure safety, reduce accidental border crossings, and improve overall fishing confidence and sustainability.

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CHAPTER 1

INTRODUCTION

The Indian coastline is not only a geographic boundary but also a lifeline for thousands of coastal communities whose lives and livelihoods are deeply entwined with the ocean. Among these communities, the fishermen of Rameshwaram — a historic and strategic town located on the southeastern tip of India — face a unique set of challenges. One of the most critical and persistent threats they encounter is the inadvertent crossing of the **International Maritime Boundary Line (IMBL)** that separates Indian waters from Sri Lankan territory. These accidental transgressions, often a result of poor visibility, lack of awareness, or inadequate navigational tools, frequently lead to severe consequences such as arrest, imprisonment, confiscation of boats, and deep financial loss. The cumulative impact of these incidents not only endangers the safety of fishermen but also sows fear and uncertainty within entire coastal communities.

In response to this pressing issue, we present an innovative, user-centered solution — a **LoRa-based mobile application** designed specifically for Rameshwaram fishermen. This project, titled *"Navigating the Divide,"* leverages **LoRa (Long Range)** communication technology to bridge the critical information gap between the fishermen and maritime boundaries. Our solution focuses on delivering **real-time location tracking, proximity alerts, and visual cues** regarding the IMBL, even in low-connectivity environments. With features such as GPS integration, offline map support, and an intuitive interface tailored for low-tech users, the app aims to empower fishermen with the knowledge they need to stay safe .

Through detailed research into the lives and struggles of local fishermen — particularly those with minimal technological exposure — we have created a user-friendly, effective system rooted in empathy and practical design.; it is a step toward restoring confidence, ensuring safety, and preserving the livelihoods of a marginalized community. As we move toward real-world testing and deployment, we envision a future where no fisherman has to risk his freedom for a better catch — where technology serves as both a guide and a guardian in the vast and unpredictable waters they call home.

1.1 Motivation

Enhancing Fishermen Safety: Rameshwaram fishermen often face risks of unintentional border crossings into Sri Lankan waters, resulting in arrests and financial losses. This project aims to enhance their safety by providing real-time alerts near the International Maritime Boundary Line (IMBL).

Protecting Livelihoods: By reducing accidental boundary violations, the app helps protect the livelihoods of fishermen, allowing them to focus on their work without fear of crossing into restricted zones.

Leveraging LoRa Technology: The project utilizes LoRa communication for reliable, long-range connectivity in offshore environments. This enables real-time GPS tracking and alert systems, even in areas with limited cellular coverage.

1.2 Objectives

- 2 • **Prevent Unintentional Border Crossings:**
Develop a system that alerts fishermen when approaching the International Maritime Boundary Line (IMBL) to reduce accidental entry into Sri Lankan waters.
- 3 • **Enhance Maritime Safety:**
Improve the overall safety of fishermen by providing real-time location tracking and proximity alerts while at sea.
- 4 • **Develop a User-Friendly Mobile Application:**
Create an intuitive and accessible interface tailored for fishermen with limited technological literacy.
- 5 • **Utilize LoRa for Offshore Communication:**
Implement LoRa technology to enable long-range, low-power communication where traditional mobile networks are unavailable.
- 6 • **Support Offline Functionality:**
Incorporate GPS and offline map features to ensure the app functions reliably in remote maritime areas

CHAPTER 2

LITERATURE REVIEW

1. **Challenges in Maritime Boundary Navigation:**

Numerous studies have highlighted the risks faced by Indian fishermen who unknowingly cross the International Maritime Boundary Line (IMBL) due to lack of awareness or accurate navigational tools. According to reports, this often results in detainment by foreign authorities, seizure of boats, and long-term legal struggles. These challenges underscore the need for reliable and real-time alert systems.

2. **Existing Marine Tracking Systems:**

Traditional marine tracking technologies such as AIS (Automatic Identification System) are commonly used by commercial vessels but are either too costly or complex for small-scale fishermen. Moreover, AIS requires continuous internet connectivity, which is not always feasible in remote offshore areas.

3. **Use of LoRa in Marine and Remote Applications:**

LoRa (Long Range) technology has been gaining attention for its potential in low-power, long-distance communication. Research has shown its effectiveness in remote environmental monitoring, smart agriculture, and disaster alert systems. Its ability to transmit over several kilometers without internet makes it ideal for maritime communication.

4. **Mobile Applications for Safety and Navigation:**

There are mobile apps designed for navigation and marine safety; however, few are tailored to low-literacy users or function offline. Research recommends user-centric designs, particularly for vulnerable populations, to ensure usability and effectiveness.

5. **Gaps and Opportunities**

While LoRa-based systems have been explored in agriculture and urban applications, their use in coastal fishing communities remains limited. This presents an opportunity to innovate a specialized, scalable solution that bridges the gap between technology and real-world safety needs.

2.1 Existing System

2.1.1 Advantages of the existing system

- 3 Rameshwaram fishermen currently depend on traditional navigation methods like visual landmarks, stars, and basic GPS devices to guide their movements at sea. While these methods help with direction, they do not provide any information or warnings about the International Maritime Boundary Line (IMBL). Some boats are equipped with GPS units, but these require constant monitoring and do not offer automated alerts.
- 4 Advanced maritime tracking systems like AIS (Automatic Identification System) are available, but they are costly, complex, and primarily used by commercial or government vessels—making them unsuitable for small-scale fishermen. Additionally, a few mobile applications for marine navigation exist, but most depend heavily on internet or cellular connectivity, which is often unavailable or unstable in remote offshore regions.
- 5 As a result, the existing systems fail to meet the practical needs of local fishermen for safe, real-time boundary detection and alerts.

Advantages of existing system :

- ☐ **Familiarity:**
Traditional navigation methods are well-known and trusted by local fishermen.
- ☐ **Low Initial Cost:**
Using basic tools like compasses or simple GPS devices requires little to no investment.
- ☐ **No Technical Training Required:**
Traditional methods and basic GPS use do not demand advanced technological skills.
- ☐ **Independence from Network Infrastructure:**
Basic GPS systems function without the need for mobile or internet connectivity

5.1.1 Drawbacks of the existing system

1. **No IMBL Alerts:**
Traditional methods and basic GPS do not provide warnings when nearing the International Maritime Boundary Line.
2. **Risk of Unintentional Border Crossing:**
Without boundary awareness, fishermen may unknowingly enter Sri Lankan waters.
3. **Lack of Real-Time Monitoring:**
No dynamic tracking or automatic alert system to help with decision-making at sea.
4. **Limited Accuracy and Reliability:**
Traditional navigation is prone to human error, especially in bad weather or low visibility.

5.2 Proposed System

The proposed system is a **LoRa-based mobile application** specifically designed to assist Rameshwaram fishermen in safely navigating near the International Maritime Boundary Line (IMBL). It integrates **GPS tracking, real-time proximity alerts, and offline map support** to ensure accurate and timely warnings even in areas with poor or no internet connectivity.

The system uses **LoRa (Long Range) communication technology**, which enables low-power, long-distance data transmission suitable for offshore conditions. When a fishing boat approaches within 50 nautical miles of the IMBL, the system automatically triggers **visual and audio alerts** to warn the user. The mobile app features a **simple, intuitive interface** that is easy to use even by fishermen with minimal tech literacy.

The goal of the proposed system is to **prevent unintentional IMBL crossings**, improve **safety at sea**, and **protect the livelihoods** of fishermen by avoiding legal issues and boat seizures.

5.2.1 Advantages of the proposed system

Improves safety by alerting fishermen before they near the IMBL.

Works without mobile networks using long-range LoRa technology.

Easy to use, even for those with basic smartphone skills.

Supports offline maps for uninterrupted use at sea.

Low-cost and scalable for wider coastal deployment.

Protects livelihoods by preventing arrests and boat seizures.

CHAPTER 3

SYSTEM DESIGN

3.1 Development Environment

3.1.1 Hardware Requirements

Arduino UNO

Bread Board

Gps module

Jumper wires

lora

transmitter

and

receiver, buzzer

r

Phone ,waterproof enclosure

GPS Module

- Used to obtain accurate, real-time location coordinates of the fishing boat.
- Essential for detecting proximity to the International Marine Boundary Line (IMBL).

LoRa Transmitter and Receiver (e.g., SX1278)

- Enables long-range, low-power wireless communication between the boat and the onshore base station.
- Works efficiently in offshore areas without internet or mobile network coverage.

Microcontroller (Arduino Uno)

- Acts as the control unit to process GPS data, manage LoRa communication, and trigger alerts.
- Chosen for its compact design, low power consumption, and support for wireless modules.

Power Source (Battery Pack or Solar Panel)

- Provides uninterrupted power to all onboard electronics.
- Solar panels are ideal for long-duration use at sea.

Buzzer or Vibration Motor

- Delivers immediate physical alerts (sound or vibration) to the fisherman when approaching restricted waters.
- Ensures alerts are received even if the mobile app is not actively monitored.

Mobile Device (Smartphone or Tablet)

- Runs the user-facing app that displays maps, alerts, and location status.

- Must support basic GPS and app functionality with offline capability.

Waterproof Enclosure

- Protects electronic components from harsh marine environments like saltwater, moisture, and physical shocks.

3.1.1 Software Requirements

- ☐ Mobile Application (Client App)
 - Platform: Android (can also be extended to iOS)
 - Framework: Flutter (cross-platform compatibility)
 - Functions:
 - Display real-time location on a map
 - Show proximity alerts to the IMBL
 - Work offline using preloaded maps
 - Provide visual and audio alerts to the fisherman
- ☐ Server-Side Application (Admin Dashboard)
 - Platform: Web-based (can also be accessed from desktop or mobile)
 - Technologies: Firebase (or alternative backend), Folium for map rendering, FTP for data transfer
 - Functions:
 - Monitor multiple boats in real time
 - Log data for verification and compliance
 - Manage alert history and boundary settings
- ☐ GPS Data Processing Algorithm
 - Compares real-time coordinates with predefined maritime boundaries
 - Triggers alerts when the boat approaches or crosses a threshold (e.g., 50 nautical miles)
- ☐ Geospatial Database
 - Stores IMBL boundary data and historical movement records
 - Used for alert verification and dispute resolution
- ☐ LoRa Communication Protocol
 - Ensures structured transmission of GPS coordinates from boat to shore
 - Can operate without internet, using point-to-point or LoRaWAN network
- ☐ Alert Notification System
 - Sends immediate alerts to both the mobile device and onboard buzzer
 - Supports audio, visual, and vibration-based alerts for redundancy
- ☐ Development Tools / IDEs
 - Visual Studio Code or Android Studio for app development
 - Arduino IDE for microcontroller programming

CHAPTER 4

PROJECT DESCRIPTION

4.1 ARCHITECTURE DIAGRAM

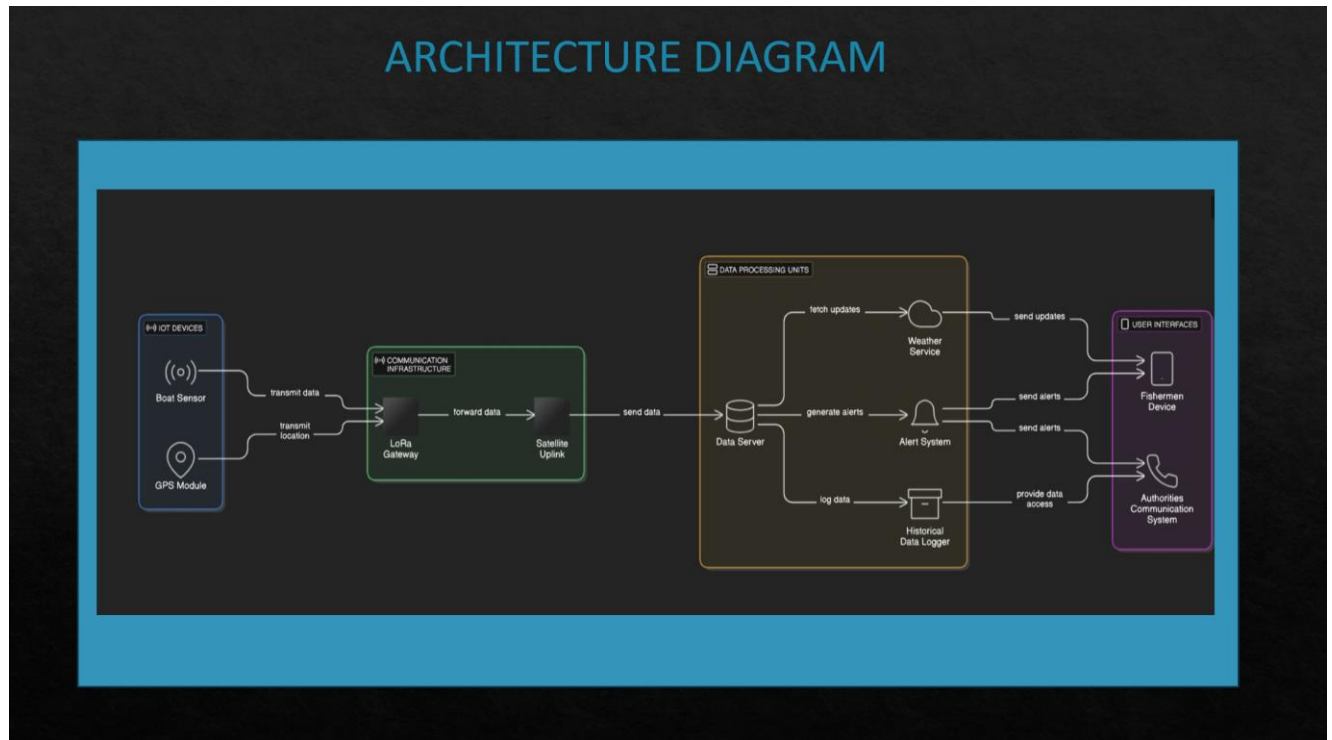


Fig 4.1 Architecture system

4.2 METHODOLOGY

Problem Identification

- Recognize the issue of accidental IMBL crossings by fishermen near Rameshwaram.
- Analyze current limitations like lack of real-time monitoring and network connectivity at sea.

Requirement Analysis

- Understand user needs (fishermen) through empathy mapping and personas.
- Define functional and non-functional requirements for the system (e.g., offline capability, instant alerts).

System Design

- Design the system architecture integrating GPS modules, LoRa transceivers, and a mobile app.
- Establish communication flow between fishing boat (client side) and onshore base

station (server side).

- Use geospatial boundary data to create a virtual buffer zone around the IMBL.

Hardware Integration

- Assemble components: GPS module, LoRa module, ESP32/Arduino microcontroller, buzzer, power supply.
- Program microcontroller to read GPS data and transmit it via LoRa.

Software Development

- Develop a mobile application using Flutter for cross-platform compatibility.
- Implement real-time map display, alert generation, and offline operation features.
- Create a server-side application using Firebase, Folium, and geospatial processing tools for monitoring and data logging.

Boundary Detection & Alert Logic

- Continuously track the boat's location via GPS.
- Calculate the distance to the IMBL.
- Trigger alerts when the boat enters a pre-defined alert zone (e.g., within 50 nautical miles of the IMBL).

Testing & Validation

- Test the complete system in controlled environments and simulated conditions.
- Validate GPS accuracy, LoRa communication reliability, and alert responsiveness.

Deployment & Feedback

- Deploy the system on actual fishing vessels in Rameshwaram.
- Collect user feedback for further refinement and enhancement.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The proposed LoRa-based Maritime Border Alert System offers an innovative and practical solution to a critical real-world problem faced by fishermen near international maritime boundaries. By integrating GPS tracking, LoRa communication, and real-time alert mechanisms, the system ensures the safety and legal compliance of fishermen operating in remote offshore areas. Its offline functionality, user-friendly interface, and low-cost hardware make it highly effective and scalable. This project not only enhances maritime security but also empowers local fishing communities by protecting their livelihoods and preventing unintended cross-border incidents.

Future Work

- Integrate with government surveillance for coordinated response.
- Add voice alerts in local languages for better accessibility.
- Incorporate satellite backup for extended range.
- Include weather and sea condition warnings.
- Explore wearable alert devices for ease of use.

APPENDIX

SOFTWARE INSTALLATION

Arduino IDE

To run and mount code on the Arduino NANO, we need to first install the Arduino IDE. After running the code successfully, mount it.

Sample code

```
#include <LiquidCrystal.h>
#include <SoftwareSerial.h>
```

```
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
SoftwareSerial gpsSerial(10, 9); // RX, TX for GPS
```

```
const int buzzerPin = 7;
```

```
// Simulated test coordinates (for demo without GPS)
float latitude = 9.2876;
float longitude = 79.3172;
```

```
void setup() {
  lcd.begin(16, 2);
  Serial.begin(9600);
  gpsSerial.begin(9600);
  pinMode(buzzerPin, OUTPUT);
  lcd.print("Initializing...");
  delay(2000);
  lcd.clear();
}
```

```
void loop() {
  // Simulate GPS values for testing boundary detection
  // You can replace this with real-time GPS parsing later
  latitude += 0.05; // Simulating movement
  longitude += 0.05;

  lcd.setCursor(0, 0);
  lcd.print("Lat:");
  lcd.print(latitude, 2);

  lcd.setCursor(0, 1);
  lcd.print("Lon:");
```

```

lcd.print(longitude, 2);

Serial.print("Latitude: "); Serial.println(latitude, 6);
Serial.print("Longitude: "); Serial.println(longitude, 6);

if (isInRestrictedZone(latitude, longitude)) {
    Serial.println("ALERT! Boundary Crossed");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("*** ALERT ***");
    lcd.setCursor(0, 1);
    lcd.print("Boundary Crossed");

    digitalWrite(buzzerPin, HIGH);
    delay(2000);
    digitalWrite(buzzerPin, LOW);
}

delay(3000); // Wait before next update
lcd.clear();
}

// Simple function to define boundary zone
bool isInRestrictedZone(float lat, float lon) {
    return (lat > 9.5 || lon > 80.0); // Sri Lanka boundary (adjust as needed)
}

```

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